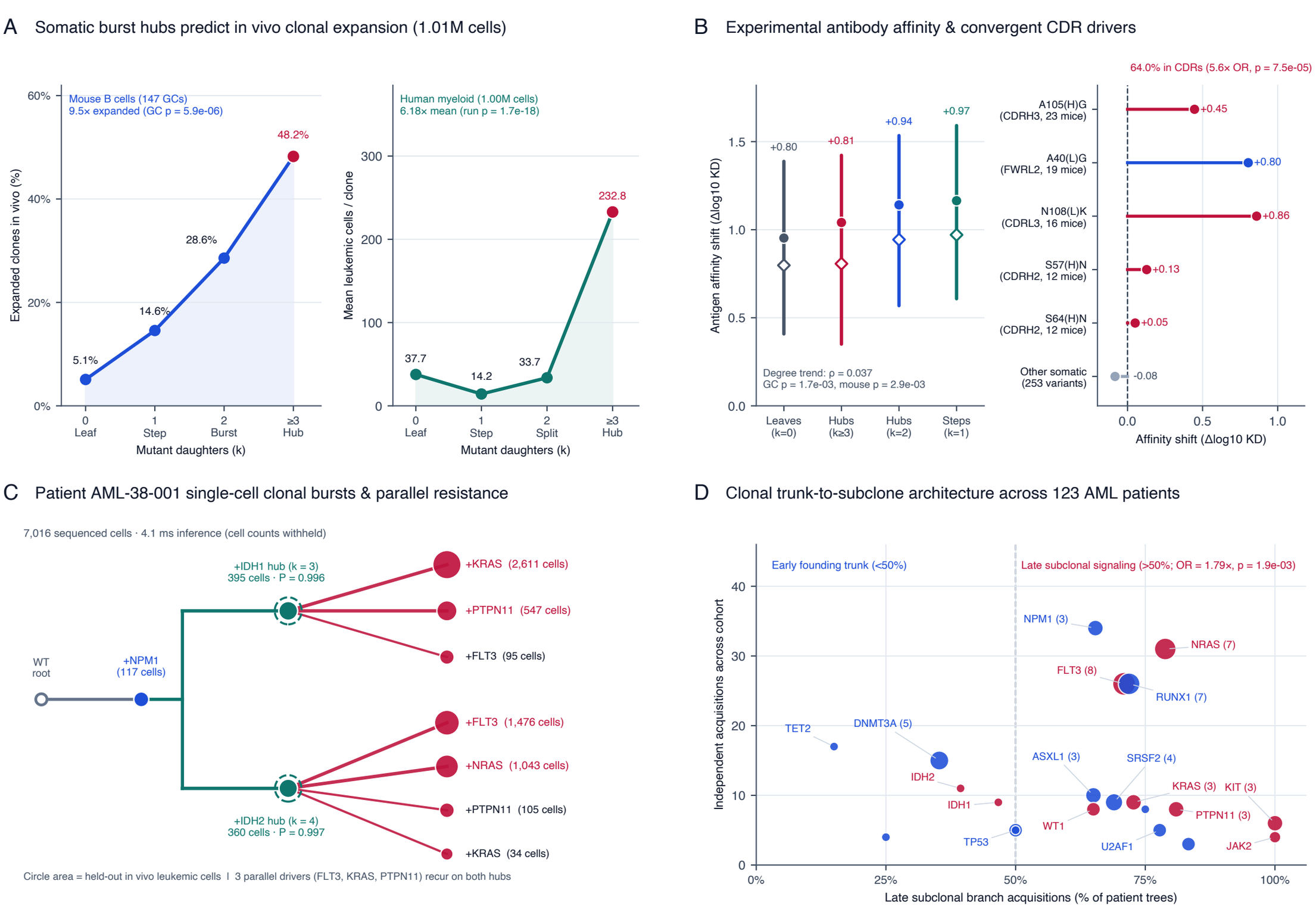
A
Somatic burst hubs predict in vivo clonal expansion (1.01M cells)
Mouse B cells (147 GCs)
9.5× expanded (GC p = 5.9e-06)
Expanded clones in vivo (%)
60%
40%
20%
0%
5.1%
14.6%
28.6%
48.2%
0 Leaf
1 Step
2 Burst
≥3 Hub
Mutant daughters (k)
Human myeloid (1.00M cells)
6.18× mean (run p = 1.7e-18)
Mean leukemic cells / clone
300
200
100
0
37.7
14.2
33.7
232.8
0 Leaf
1 Step
2 Split
≥3 Hub
Mutant daughters (k)
B
Experimental antibody affinity & convergent CDR drivers
Antigen affinity shift (Δlog10 KD)
1.5
1.0
0.5
0.0
+0.80
+0.81
+0.94
+0.97
Degree trend: ρ = 0.037
GC p = 1.7e-03, mouse p = 2.9e-03
Leaves (k=0)
Hubs (k≥3)
Hubs (k=2)
Steps (k=1)
64.0% in CDRs (5.6× OR, p = 7.5e-05)
A105(H)G (CDRH3, 23 mice)
+0.45
A40(L)G (FWRL2, 19 mice)
+0.80
N108(L)K (CDRL3, 16 mice)
+0.86
S57(H)N (CDRH2, 12 mice)
+0.13
S64(H)N (CDRH2, 12 mice)
+0.05
Other somatic (253 variants)
-0.08
0.0
0.5
1.0
Affinity shift (Δlog10 KD)
C
Patient AML-38-001 single-cell clonal bursts & parallel resistance
7,016 sequenced cells · 4.1 ms inference (cell counts withheld)
WT root
+NPM1 (117 cells)
+IDH1 hub (k = 3)
395 cells · P = 0.996
+KRAS (2,611 cells)
+PTPN11 (547 cells)
+FLT3 (95 cells)
+FLT3 (1,476 cells)
+NRAS (1,043 cells)
+PTPN11 (105 cells)
+IDH2 hub (k = 4)
360 cells · P = 0.997
+KRAS (34 cells)
Circle area = held-out in vivo leukemic cells I 3 parallel drivers (FLT3, KRAS, PTPN11) recur on both hubs
D
Clonal trunk-to-subclone architecture across 123 AML patients
Early founding trunk (<50%)
Late subclonal signaling (>50%; OR = 1.79×, p = 1.9e-03)
Independent acquisitions across cohort
40
30
20
10
0
NPM1 (3)
NRAS (7)
FLT3 (8)
RUNX1 (7)
TET2
DNMT3A (5)
ASXL1 (3)
SRSF2 (4)
IDH2
KRAS (3)
KIT (3)
IDH1
PTPN11 (3)
WT1
TP53
U2AF1
JAK2
0%
25%
50%
75%
100%
Late subclonal branch acquisitions (% of patient trees)